Diversifying Seeds and Audience in Social Influence Maximization

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***Abstract*—Influence maximization (IM) has been extensively studied for better viral marketing. However, previous works put less emphasis on how balancedly the audience are affected across different communities and how diversely the seed nodes are selected. In this paper, we incorporate audience diversity and seed diversity into the IM task. From the model perspective, in order to characterize both influence spread and diversity in our objective function, we adopt three commonly used utilities in economics (i.e., Perfect Substitutes, Perfect Complements and Cobb-Douglas). We validate our choices of these three functions by showing their nice properties. From the algorithmic perspective, we present various approximation strategies to maximize the utilities. In audience diversification, we propose a solution-dependent approximation algorithm to circumvent the hardness results. In seed diversification, we prove a $(1/e - \epsilon)$ approximation ratio based on non-monotonic submodular maximization. Experimental results show that our framework outperforms other natural heuristics both in utility maximization and result diversification.**

I. Introduction

Viral marketing through social networks is becoming more and more popular due to the rapid growth of social media. Many advertisers promote their products by paying influential users, hoping the advertisement can propagate through reposts or shares. Kempe et al. [10] first formulate this task as Influence Maximization (IM), in which they aim to obtain the largest influence spread with at most k seed nodes.

Although IM and many of its variants have been extensively studied, most approaches disregard another concern – *diversity of nodes*. Diversity has been regarded as a crucial factor in many other tasks (e.g., document retrieval [4] and node ranking [9]). These tasks mainly focus on the diversity of selected items. In contrast, in IM, we argue that both the diversity of *selected nodes* (i.e., seeds) and the diversity of *activated nodes* (i.e., audience) need to be explored.

Audience Diversification. In practical viral marketing scenarios, having diverse audience could bring many benefits. As we all know, a network contains multiple communities, and people in different communities may have different probabilities to buy the product after they see the advertisement. As mentioned in [18], if all targeted users come from one group or share the same feature, we are actually "putting all eggs in one basket", which will definitely increase the risk of marketing campaigns.

Seed Diversification. Inspired by the widely observed *homophily phenomenon* [13], [15], Tang et al. [18] relax the problem from enforcing diversity on influenced crowd to just enforcing diversity on seed set. The idea behind this relaxation is that nodes with similar features are more likely to connect with each other. Therefore, if the seed set is diverse, the activated audience would be diverse as well. Besides, some real applications directly require the diversity of seed nodes. Consider the task of setting up a conference programming committee [14]. The goal is to invite influential researchers from all related areas. Without diversity, the interest of the committee could be biased.

In this paper, we study *Audience-Diversified IM* (ADIM) and *Seed-Diversified IM* (SDIM) in a unified framework. In ADIM, we model influence spread in each community as a factor in the objective function. Intuitively, the objective should satisfy two properties: (1) *Pareto Efficiency*: Activating more nodes in one community while keeping the spreading results in other communities will increase the objective value, and (2) *Community-Level Balance*: Fixing the total influence spread, a more equally distributed spreading result over the communities will have a higher objective value. In SDIM, we model overall spread and seed diversity as two factors in our objective. Again, the selection of objectives should follow two principles: (1) *Pareto Efficiency*: Increasing either spread or diversity will boost the objective value, and (2) *Spread-Diversity Tradeoff*: When diversity is lower, we are more willing to substitute spread for diversity.

Inspired by economics practice [22], we propose to use three kinds of utility functions, *Perfect Substitutes*, *Perfect Complements* and *Cobb-Douglas* to composite multiple factors into the final objective. We first prove that the three utility functions all satisfy the properties mentioned above. Then we propose algorithms to maximize these objectives case by case. In ADIM, we prove that for Perfect Substitutes, the hill-climbing greedy method can provide a $(1 - 1/e - \epsilon)$ approximation guarantee. For Perfect Complements and Cobb-Douglas, we show that ADIM is hard to approximate with any positive constant ratio. Given the hardness, we propose an algorithm with solution-dependent approximation guarantees using the *Sandwich Approximation* strategy [5], [11], [12]. In SDIM, we prove the three utilities are all submodular (but not monotonic). Therefore, the *Random Greedy* strategy [3] for size-constrained non-monotonic submodular optimization can guarantee a $(1/e - \epsilon)$ approximation ratio.

We conduct extensive experiments on two real-world networks with different choices of utility functions. Our method consistently outperforms several baselines including traditional IM [10], diversified node ranking [9] and diversified IM [18] in both ADIM and SDIM. Besides, putting the utilities aside, we prove the success of our diversified IM framework from the views of *entropy* [18] and *coverage* [25], indicating that the result is truly diversified, not just maximizing an objective.

II. Problem Formulation

A. Audience-Diversified Influence Maximization (ADIM)

Influence Maximization (IM) [10]. In a network $G = (V, E)$, an information cascade starts from an initially active set $S \subseteq V$. Then the information propagates in G under certain spreading models as the time runs forward. There are two spreading models [10] commonly studied in IM.

In the *Independent Cascade (IC)* model, when a node u becomes active at time t, it gets a chance to activate each of its inactive neighbor v at time $t+1$, with probability p_{uv}. If v is not influenced by u, u cannot make any further attempts in the subsequent rounds.

In the *Linear Threshold (LT)* model, each node v has a threshold $\theta_v \sim U[0,1]$ and each edge (u, v) has a weight b_{uv}. If at time t, we have $\sum_{v\text{'s active neighbor } u} b_{uv} \geq \theta_v$ for an inactive node v, then v will become active at time $t+1$.

Given the spreading model, the expected influence spread is defines as

$$\sigma(S) = \mathbb{E}[|S_{\texttt{act}}|],$$

where $S_{\texttt{act}}$ is the set of nodes activated in the cascade (including S). IM aims to find an S with at most k nodes to maximize $\sigma(S)$.

Audience Diversity. To introduce audience diversity, we assume there are C communities (i.e., audience groups) $V_1, V_2, ..., V_C$ in a network, where V_c is the set of nodes in the c-th community. Similar to IM, the expected influence spread in V_c is

$$\sigma_c(S) = \mathbb{E}[|S_{\texttt{act}} \cap V_c|].$$

Considering influence spread in each community as a factor, the objective function can be defined as

$$f(S) = F\Big(\alpha_1\sigma_1(S), \alpha_2\sigma_2(S), ..., \alpha_C\sigma_C(S)\Big),$$

where α_c is the weight of community V_c in the utility. For example, if we would like to study the "proportion" of activated nodes in each community instead of the absolute number, we can set $\alpha_c = 1/|V_c|$.

Intuitively, F should satisfy the following two properties:

(P1) Pareto Efficiency. $\forall c$ and $\epsilon \geq 0$, $F(x_1, ..., x_{c-1}, x_c, x_{c+1}, ..., x_C) \leq F(x_1, ..., x_{c-1}, x_c + \epsilon, x_{c+1}, ..., x_C)$.

(P1) models the fact that we favor an action which benefits at least one community without making the others worse off.

(P2) Community-Level Balance. $\forall k$ $(1 \leq k \leq C)$ communities (w.l.o.g., $V_1, ..., V_k$), $F(x_1, ..., x_k, x_{k+1}, ...x_C) \leq F(\frac{x_1+...+x_k}{k}, ..., \frac{x_1+...+x_k}{k}, x_{k+1}, ..., x_C)$.

(P2) models our preference for a more equally distributed (i.e., diversified) spreading result over communities.

There are many candidates satisfying (P1) and (P2). In this paper, inspired by the utility functions in economics, we focus on three cases which are commonly adopted to characterize users' preference when there are multiple factors [22].

Perfect Substitutes (Linear Utility). Two goods are perfect substitutes if the user is willing to substitute one good for the other at a constant rate (e.g., for most people, Pepsi and Coke). In mathematics, the function is essentially a weighted sum.

$$f_S(S) = \sum_{c=1}^{C} \alpha_c\sigma_c(S), \qquad (\alpha_c > 0).$$

Since $f_S(\cdot)$ is very similar to the global spread $\sigma(\cdot)$, we consider a more generalized form called *Constant Elasticity of Substitution* (CES) [22].

$$f_{CES}(S) = \Big(\sum_{c=1}^{C}(\alpha_c\sigma_c(S))^{\rho}\Big)^{1/\rho}, \quad (0 < \rho \leq 1,\ \alpha_c > 0).$$

Perfect Complements (Leontief Utility). A nice example is that of left shoes and right shoes. if we have exactly two pairs of shoes, then neither extra left shoes nor extra right shoes can do us a bit of good. When the influence spread in different communities are regarded as perfect complements, we have

$$f_C(S) = \min_{1\leq c\leq C} \alpha_c\sigma_c(S), \qquad (\alpha_c > 0).$$

Cobb-Douglas Utility. In economics, the utility function usually follows the law of diminishing returns: Adding more of one factor, while holding all other constant, will yield lower incremental per-unit returns. Cobb-Douglas utility is commonly used to describe this property:

$$f_D(S) = \prod_{c=1}^{C}(\alpha_c\sigma_c(S))^{1/C}, \qquad (\alpha_c > 0).$$

The three kinds of utilities correspond to $F_S(x) = \sum_c x_c$ (more generally, $F_{CES}(x) = (\sum_c x_c^{\rho})^{1/\rho}$), $F_C(x) = \min_c x_c$ and $F_D(x) = (\prod_c x_c)^{1/C}$, respectively.

Theorem 1. *F_{CES} $(0 < \rho \leq 1)$, F_C and F_D all satisfy (P1) and (P2).*[1]

Definition 1 (ADIM). *In a network $G = (V, E)$, given a size constraint k and a utility function $f \in \{f_{CES}$ $(0 < \rho \leq 1), f_C, f_D\}$, $\max_{|S|=k} f(S)$.*

B. Seed-Diversified Influence Maximization (SDIM)

Audience diversity can be *implicitly* described by the spreading results in all communities. In contrast, it is difficult to characterize seed diversity reversely using influence spread. Therefore, we present an *explicit* definition of seed diversity.

Seed Diversity. Assume we have a pairwise node similarity function $\text{Sim}(\cdot,\cdot) \in [0, 1]$. Inspired by the studies in diversified ranking [2], [8], [9], we define seed diversity $d(S)$ as the average pairwise dissimilarity in S.

$$d(S) = \frac{1}{|S|(|S|-1)} \sum_{u,v\in S, u\neq v} \Big(1 - \text{Sim}(u, v)\Big).$$

The form of $\text{Sim}(\cdot,\cdot)$ can be specified from various perspectives (e.g., node embeddings [19], types, etc.). Following the setting in ADIM, we partition the network into C communities. If the communities are *overlapping*, BigCLAM [23] uses a vector $F_u = [F_{u1}, ..., F_{uC}]^T$ to represent node u, where F_{uc}

[1]Please refer to our full version [24] for proofs of all the theorems.

is the probability that u belongs to community V_c. In [23], the proximity between u and v is defined as

$$\text{Sim}_C(u, v) = 1 - \exp(-F_u^T F_v).$$

The same formula can be adopted for *disjoint* communities (in which F_u becomes a "one-hot" vector), and we will have $\text{Sim}_C(u, v) = 1 - 1/e$ if u and v belong to the same community, and 0 otherwise.

We would like to mention that our dissimilarity function $1 - \text{Sim}(\cdot, \cdot)$ *need not* be a metric. For example, it is easy to see that $1 - \text{Sim}_C(u, u) \not\equiv 0$.

Following ADIM, we define an objective function that jointly models influence spread and seed diversity as two factors.

$$g(S) = G\Big(\sigma(S),\ \beta \cdot d(S)\Big),$$

where β is a constant factor making $\beta \cdot d(S)$ share the same magnitude with $\sigma(S)$ (e.g., $\beta = |V|$).

Intuitively, G should also satisfy (P1) with two variables. Besides, we propose the following property to characterize our willingness to substitute spread for diversity.

(P3) Spread-Diversity Tradeoff. If $\exists\ \epsilon, \delta \geq 0$ such that $G(x_1 - \epsilon, x_2 + \delta) = G(x_1, x_2)$, then $G(x_1 - 2\epsilon, x_2 + 2\delta) \leq G(x_1 - \epsilon, x_2 + \delta)$.

(P3) tells us that when diversity is lower (i.e., x_2), we are more willing to substitute spread for diversity (i.e., at the rate of ϵ/δ). When diversity becomes higher (i.e., $x_2 + \delta$), we no longer expect the substitution at the same rate. In economics, this is named as the law of diminishing marginal rates of substitution [22].

We can still adopt the three utility functions used in ADIM.

Perfect Substitutes. When $G_S(x_1, x_2) = x_1 + x_2$, we have

$$g_S(S) = \sigma(S) + \beta \cdot d(S), \qquad (\beta > 0).$$

Essentially, Perfect Substitutes is a weighted sum of spread and diversity. [8] and [2] also studied this utility function for diversified ranking. However, in their models, $1 - \text{Sim}(\cdot, \cdot)$ must form a metric. Without this assumption (e.g., in our community and embedding settings), their algorithms do not have approximation guarantees.

Perfect Complements. When $G_C(x_1, x_2) = \min\{x_1, x_2\}$,

$$g_C(S) = \min\{\sigma(S),\ \beta \cdot d(S)\}, \qquad (\beta > 0).$$

Cobb-Douglas Utility. When $G_D(x_1, x_2) = x_1^a x_2^b$ ($0 < a, b \leq 1$ and $a + b = 1$), we have

$$g_D(S) = \sigma(S)^a \cdot (\beta d(S))^b \propto \sigma(S)^a \cdot d(S)^b.$$

Theorem 2. *G_S, G_C and G_D all satisfy (P1) and (P3).*

Note that (P2) and (P3) are not equivalent. For example, $x_1^{0.9} x_2^{0.1}$ satisfies (P3) but violates (P2).

Definition 2 (SDIM). *In a network $G = (V, E)$, given a size constraint k and a utility function $g \in \{g_S, g_C, g_D\}$, $\max_{|S|=k} g(S)$.*

It is easy to show that both ADIM and SDIM can be viewed as the extension of traditional IM, so they are NP-hard.

TABLE I
APPROXIMATION RESULTS FOR ADIM AND SDIM.

Utility	Audience Diversification	Seed Diversification
Substitutes	$1 - 1/e - \epsilon$ $(1 - 1/e - \epsilon)^{1/\rho}$ for CES	$1/e - \epsilon$
Complements	NP-Hard to Approx. Solution-dependent Approx.	$1/e - \epsilon$
Cobb-Douglas	NP-Hard to Approx. Solution-dependent Approx.	$1/e - \epsilon$

III. ALGORITHMS

Due to NP-hardness, we focus on finding approximation algorithms for ADIM and SDIM. Table I summarizes our results in this section.

A. Audience-Diversified Influence Maximization

Perfect Substitutes and CES. With the help of submodularity, Perfect Substitutes and CES ($0 < \rho \leq 1$) can be solved via the traditional hill-climbing greedy method.

Lemma 1 [18]. *Under IC or LT model, $\sigma_c(S)$ is monotonic and submodular for any $c = 1, ..., C$.*

Theorem 3. *Under IC or LT model,* GREEDY*(f_{CES}^{ρ}, k) achieves a $(1 - 1/e - \epsilon)^{1/\rho}$ approximation guarantee when $0 < \rho \leq 1$.*

When $\rho = 1$, we have the common $(1 - 1/e - \epsilon)$ approximation ratio for Perfect Substitutes.

Perfect Complements. Algorithm 1 cannot be applied to $f_C(\cdot)$ and $f_D(\cdot)$ because neither of them is submodular. In fact, it is hard to obtain any positive constant approximation guarantee in these two cases.

Theorem 4. *Under IC model, ADIM is NP-hard to approximate with any positive constant factor for $f_C(\cdot)$ and $f_D(\cdot)$.*

To circumvent this hardness result, we attempt to prove a *solution-dependent* guarantee [11], [12]. In light of the *Sandwich Approximation* (SA) strategy [12], we propose Algorithm 2 that works for both $f_C(\cdot)$ and $f_D(\cdot)$.

SA aims to optimize a submodular upper bound of the original objective (or a lower bound [11], or both [12]). To be specific, we look for a submodular function f_C^+, where $f_C(S)$ is always smaller than $f_C^+(S)$. In the case of Perfect Complements, there are C upper bounds $\alpha_i \sigma_i(S)$ ($i = 1, 2, ..., C$), each of which is monotonic and submodular. When we apply the SA strategy on all of these C upper bounds, the following result can be derived.

Theorem 5. *Under IC or LT model,* UPPER-GREEDY*(f_C, $\{\alpha_1\sigma_1, ..., \alpha_C\sigma_C\}, k$) finds a seed set S and guarantees that*

$$f_C(S) \geq \max_{1 \leq i \leq C} \frac{f_C(S_i)}{\alpha_i \sigma_i(S_i)} (1 - \frac{1}{e} - \epsilon) f_C(S_C^*),$$

where S_C^* is the optimal solution for $f_C(\cdot)$.

$\max_{1 \leq i \leq C} \frac{f_C(S_i)}{\alpha_i \sigma_i(S_i)} (1 - \frac{1}{e} - \epsilon)$ is referred as a solution-dependent approximation ratio since it is related to S_i. Note that it can be calculated once we have the solution, and the true effectiveness of UPPER-GREEDY depends on the gap between $\alpha_i \sigma_i$ and f_C. In our case, when there is only one community, $f_C \equiv \alpha_1 \sigma_1$. Then Algorithm 2 has the common $(1 - 1/e - \epsilon)$ approximation ratio.

Algorithm 1 GREEDY(f, k)

```
initialize S = ∅
for i = 1 to k do
    select u = arg max_{v∈V\S}(f(S ∪ {v}) − f(S))
    S = S ∪ {u}
end for
output S
```

Algorithm 2 UPPER-GREEDY($f, \{f_1^+, ..., f_U^+\}, k$)

```
S_0 = GREEDY(f, k)
for i = 1 to U do
    S_i = GREEDY(f_i^+, k)
end for
S = arg max_{0≤i≤U} f(S_i)
output S
```

Cobb-Douglas. Again, we adopt the SA strategy. Our selection of the upper bound is $f_D^+(S) = \frac{1}{C}\sum_{c=1}^{C}\alpha_c\sigma_c(S)$. Since the geometric mean is always less than or equal to the arithmetic mean, we have $f_D^+(S) \geq f_D(S)$. Similar to Theorem 5, the following result holds.

Theorem 6. *Under IC or LT model,* UPPER-GREEDY(f_D, $\{f_{D+}\}$, k) *finds a seed set S and guarantees that*

$$f_D(S) \geq \frac{f_D(S_1)}{f_D^+(S_1)}(1 - \frac{1}{e} - \epsilon)f_D(S_D^*).$$

where S_D^* is the optimal solution for $f_D(\cdot)$.

Again, when there is only one community, f_D and f_{D+} are equivalent, and we get the common $(1-1/e-\epsilon)$ approximation ratio.

Time Complexity. The time complexity of GREEDY is $O(k|V|\mathcal{T})$, where $\mathcal{T}$ is the time to calculate $f(S)$. Chen et al. [6], [7] have pointed out the hardness of this computation under IC and LT models, but an arbitrarily small error can be obtained through Monte Carlo simulation. If we run M trials of simulation in each estimation, the overall time complexity is $O(kM|V|^2)$. Similarly, the time complexity of UPPER-GREEDY is $O(kUM|V|^2)$, where U is the number of upper bounds used. In this paper, we do not focus on the efficiency of estimating influence spread. However, it is worth noting that Monte Carlo simulations can be replaced by Reverse Influence Sampling strategies [1], [20], [21] to accelerate our algorithms.

B. Seed-Diversified Influence Maximization

Now we proceed to SDIM. Recall that $g_S(S)$, $g_C(S)$ and $g_D(S)$ are all composites of $\sigma(S)$ and $d(S)$. Although $\sigma(S)$ has good properties under IC and LT models, $d(S)$ can be neither monotonic nor submodular. To tackle this issue, we consider a problem equivalent to SDIM.

It is easy to show that

$$d(S) = 1 - \frac{1}{|S|(|S|-1)}\sum_{u,v\in S, u\neq v}\text{Sim}(u,v).$$

Now we consider

$$\tilde{d}(S) = 1 - \frac{1}{k(k-1)}\sum_{u,v\in S, u\neq v}\text{Sim}(u,v).$$

Algorithm 3 RANDOM-GREEDY(f, k)

```
initialize S_0 = ∅
for i = 1 to k do
    Let M_i ⊆ V − S_{i−1} be the subset of size k maximizing
    Σ_{v∈M_i} f(S_{i−1} ∪ {v}) − f(S_{i−1})
    Randomly select u from M_i
    S_i = S_{i−1} ∪ {u}
end for
output S_k
```

Note that $d(S) = \tilde{d}(S)$ when $|S| = k$. Therefore, for any $\tilde{g}(S) = G(\sigma(S), \tilde{d}(S))$,

$$\max_{|S|=k} g(S) \iff \max_{|S|=k}\tilde{g}(S).$$

Following this way, our SDIM problem becomes maximizing $\tilde{g}_S(S) = \sigma(S) + \beta\tilde{d}(S)$, $\tilde{g}_C(S) = \min\{\sigma(S), \beta\tilde{d}(S)\}$ and $\tilde{g}_D(S) = \sigma(S)^a\tilde{d}(S)^b$ subject to $|S| = k$.

Note that we study $\tilde{d}(S)$ instead of $d(S)$ because it has better properties.

Lemma 2. *For any Sim$(\cdot,\cdot) \in [0,1]$, $\tilde{d}(S)$ is non-negative, decreasing and submodular.*

Consequently, we can prove the following.

Theorem 7. *For any monotonic and submodular $\sigma(\cdot)$ and any Sim$(\cdot,\cdot) \in [0,1]$, $\tilde{g}_S(S)$, $\tilde{g}_C(S)$ and $\tilde{g}_D(S)$ are all non-negative and submodular.*

Theorem 7 naturally applies to $\sigma_{\text{IC/LT}}(\cdot)$ and $\text{Sim}_C(\cdot,\cdot)$. With this Theorem, we successfully transform SDIM to a size-constrained non-monotonic submodular maximization problem, where we are able to adopt the RANDOM-GREEDY algorithm (Algorithm 3) proposed in [3]. RANDOM-GREEDY is a natural generalization of the vanilla greedy algorithm. Instead of picking the best single node in each iteration, it first finds k nodes with the highest marginal gains and then randomly selects one node from the top-k candidates to add. The following result is proved in [3].

Theorem 8 [3]. *Let $g(\cdot)$ be a non-negative submodular (not necessarily monotonic) function. For the problem $\max_{|S|=k} g(S)$,* RANDOM-GREEDY(g, k) *finds a set S and guarantees $\mathbb{E}[g(S)] \geq \max\{0.266, \frac{1}{e}(1-\frac{k}{e|V|})\}\cdot g(S^*)$, where S^* is the optimal solution.*

For social influence maximization, we usually have $k = o(|V|)$. (In the real world, we can hardly obtain an initial seed set whose size is proportional to the whole network size.) In this case, the approximation rate of RANDOM-GREEDY becomes $\max\{0.266, 1/e - o(1)\} = 1/e - \epsilon$. We also assume $k \ll |V|$ in all of our experiments.

Putting Theorems 7 and 8 together, we get a $(1/e - \epsilon)$ approximation algorithm for SDIM.

Time Complexity. The time complexity of RANDOM-GREEDY is $O(k|V|\mathcal{T})$, where $\mathcal{T}$ is the time to calculate $\tilde{g}(S)$, or $\sigma(S)$ and $\tilde{d}(S)$. $\tilde{d}(S)$ can be updated incrementally in $O(k)$ time. $\sigma(S)$, again, is approximated through Monte Carlo simulation. Therefore, the overall complexity is $O(kM|V|^2)$. Reverse Influence Sampling strategies can also be applied here to devise a more efficient version of RANDOM-GREEDY.

IV. Experiments

We aim to answer three questions in our experiments: **(RQ1)** Can we achieve higher utilities in comparison with baseline algorithms? **(RQ2)** Putting the utilities aside, in ADIM, can we really diversify the activated crowd without hurting the spread? **(RQ3)** Similarly, in SDIM, can we diversify the selected seeds with little reduce in their influence power?

A. Experimental Setup

Datasets. Two benchmark networks are used: (1) FourArea [17] is an academic collaboration network extracted from DBLP. It contains authors from 4 areas: *database*, *data mining*, *machine learning* and *information retrieval*. (2) Epinions [16] is a who-trust-whom network of a consumer review site Epinions.com. We adopt BigCLAM [23] to detect 10 overlapping communities in the network. Note that in both datasets, there are nodes not belonging to any community. We summarize the dataset statistics in Table II.

TABLE II
Dataset Statistics.

Dataset	$\|V\|$	$\|E\|$	Edge Type	C	Community Type
FourArea [17]	27,199	66,832	Undirected	4	Disjoint
Epinions [16]	75,879	508,837	Directed	10	Overlapping

Algorithms. The following algorithms are involved in our comparison:

(1) **IM** [10] maximizes the spread over the whole network.

(2) **GenDeR** [9] is a generic *diversified ranking* algorithm. Here we use $\sigma(\{v\})$ as the ranking function and $\text{Sim}_C(u, v)$ as the similarity function.

(3) **Seed-DU** [18] is a *seed-diversified* IM algorithm. Following [18], we set the diversity function $f(x)$ to be $\frac{x}{1+x}$.

(4) **D-Inf** [18] is an *audience-diversified* IM algorithm. Still following [18], we set the diversity function to be $\frac{x}{1+x}$ and the balancing parameter γ to be 0.

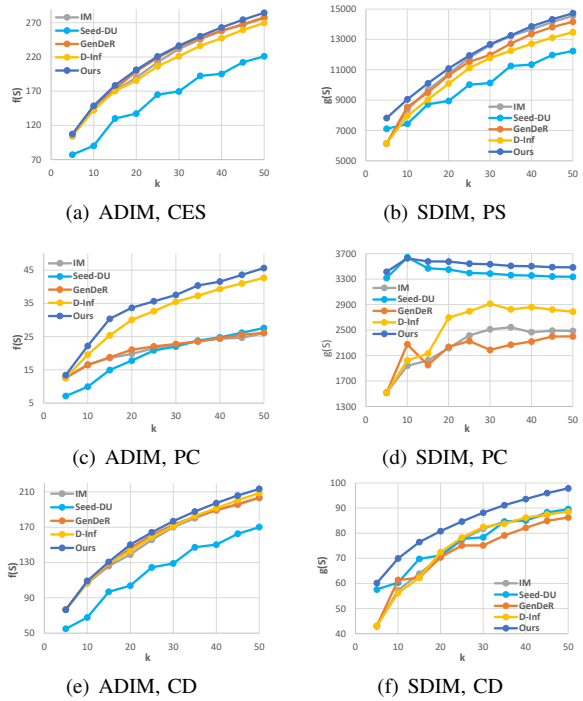
(5) **Ours** is the framework proposed in this paper.

Models and Parameters. In SDIM, we set $a = b = 1/2$ for Cobb-Douglas and $\beta = 0.05|V|$ for Perfect Substitutes and Perfect Complements. In ADIM, since Perfect Substitutes is too similar to traditional IM, we study CES ($\rho = 1/2$) instead. For Perfect Complements and Cobb-Douglas, we set $\alpha_c = 1$ ($c = 1, 2, ..., C$). For CES ($\rho = 1/2$), we set $\alpha_c = 1/C$ for normalization. We choose IC as our spreading model, where the activate probability p_{uv} is $1/\text{deg}_{\text{in}}(v)$.

B. (RQ1) Utility Maximization

ADIM. Figure 1 shows the ADIM utility values of selected nodes on Epinions. We can observe that: (1) Ours consistently performs the best with different utility functions. (2) In most cases, D-Inf performs the second best, whereas Seed-DU does not achieve satisfying utility values. This observation is aligned with their objectives. As we mentioned, D-Inf focuses on *audience diversification* while Seed-DU considers to diversify *seed nodes*. Although Tang et al. [18] use *homophily* to illustrate that diversified seeds may indicate diversified spreading results, both their experiments and ours show a gap between these two problem settings.

SDIM. Figure 1 also shows the utilities of selected nodes in SDIM. Again, Ours consistently performs the best. When



(a) ADIM, CES (b) SDIM, PS (c) ADIM, PC (d) SDIM, PC (e) ADIM, CD (f) SDIM, CD

Fig. 1. ADIM and SDIM utility values on Epinions with different utility functions (CES: CES ($\rho = 1/2$), PS: Perfect Substitutes, PC: Perfect Complements, CD: Cobb-Douglas).

using Perfect Complements and Cobb-Douglas utilities, we can outperform the baselines by a large margin; when using Perfect Substitutes, we are still the best, but the advantage against IM is slight.

We conduct the same experiments on FourArea and get similar results. One can find them in our full version [24].

C. (RQ2) Audience Diversification

Evaluation Metrics. Following [18], we define the following two metrics.

$$\text{Entropy}(S) = \sum_{i=1}^{C} -p_i \log p_i, \quad \text{where } p_i = \frac{\sigma_i(S)}{\sum_{i=1}^{C} \sigma_i(S)}.$$

$$\text{Spread}(S) = \mathbb{E}[|S_{\texttt{act}} \cap (V_1 \cup ... \cup V_C)|].$$

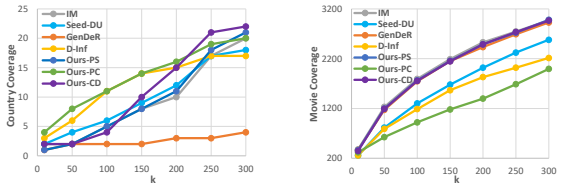
Intuitively, p_i can be interpreted as the proportion of influence distributed to community V_i, and $\text{Entropy}(S)$ reflects the degree of balance with respect to the influence spread. $\text{Spread}(S)$, from an orthogonal perspective, measures how many users *in target communities* are affected.

Results. Table III show the Entropy and Spread of cascading results on FourArea and Epinions when $k = 50$. Here "Ours-CES" means we select nodes using our approach with CES ($\rho = 1/2$). Similar meanings can be inferred for "Ours-PC" and "Ours-CD". For each algorithm, we calculate its percentage increase/decrease in comparison with IM. On the one hand, when we only focus on Entropy, Ours-PC performs the best on FourArea and the second best (and on par with the best) on Epinions. This indicates PC is the most applicable utility when users emphasize more on diversity. On the other hand, in many practical scenarios of IM, the goal is to increase the diversity of audience without hurting influence

TABLE III

ENTROPY AND SPREAD OF CASCADING RESULTS ($k = 50$) ON FOURAREA AND EPINIONS. FOR EACH ALGORITHM, WE CALCULATE ITS PERCENTAGE INCREASE/DECREASE IN COMPARISON WITH **IM**. (CES: CES ($\rho = 1/2$), PC: PERFECT COMPLEMENTS, CD: COBB-DOUGLAS.)

Dataset: FOURAREA							
Method	**IM**	**GenDeR**	**Seed-DU**	**D-Inf**	**Ours-CES**	**Ours-PC**	**Ours-CD**
Entropy	1.883	1.628	1.947	1.977	1.923	1.994	1.943
	-	**-13.5%**	**+3.4%**	**+5.0%**	**+2.1%**	**+5.8%**	**+3.2%**
Spread	344.20	338.54	315.97	339.24	354.31	256.14	350.90
	-	**-1.6%**	**-8.2%**	**-1.4%**	**+2.9%**	**-25.6%**	**+1.9%**
Dataset: EPINIONS							
Method	**IM**	**GenDeR**	**Seed-DU**	**D-Inf**	**Ours-CES**	**Ours-PC**	**Ours-CD**
Entropy	2.774	2.769	2.833	2.820	2.793	2.830	2.806
	-	**-0.2%**	**+2.1%**	**+1.7%**	**+0.7%**	**+2.0%**	**+1.2%**
Spread	3486.2	3503.9	2701.8	3305.8	3549.3	3017.0	3487.0
	-	**+0.5%**	**-22.5%**	**-5.2%**	**+1.8%**	**-13.5%**	**+0.02%**



(a) Country Coverage (b) Movie Coverage

Fig. 2. Country Coverage and Movie Coverage on the IMDB network (PS: Perfect Substitutes, PC: Perfect Complements, CD: Cobb-Douglas).

spread. Among all compared methods, only Ours-CES and Ours-CD increase Entropy and Spread simultaneously.

D. (RQ3) Seed Diversification

Following the evaluation metrics in [9], [14], [25], we conduct experiments on an actor professional network.

Dataset. The IMDB network[2] is constructed from the Internet Movie Database. Each actor/actress is represented by a node, and the edges between two nodes denote their co-starred movies. Unseen by the algorithms, each actor/actress is associated with a country.

Evaluation Metrics. Zhu et al. [25] propose two measures in a particular context of ranking movie stars, i.e., *Country Coverage* and *Movie Coverage*, which are the number of distinct countries and movies associated with the selected actors/actresses.

Results. The results are shown in Figure 2. Intuitively, Country Coverage mainly evaluates seed diversity while Movie Coverage cares more about influence power In Figure 2(b), Seed-DU, D-Inf and Ours-PC perform evidently worse, and all the other methods are on par with each other. Meanwhile, in Figure 2(a), Ours-PC and D-Inf are the best two when k is small. Without them, Ours-CD gives the most diversified results. Similar to audience diversification, we explain these observations from two perspectives. On the one hand, when we are more willing to substitute spread for diversity, Ours-PC can give us the most diversified results. On the other hand, if we would like to diversify the results with little reduce in influence power, Ours-CD is the best choice.

V. CONCLUSION

We have presented an IM framework that works for both audience diversification and seed diversification. We formulate the two tasks by carefully designing the objective to jointly describe influence spread and diversity. Three economic utilities with nice properties are adopted. Theoretically, we present various approximation algorithms to maximize the utilities. Practically, we validate the effectiveness of our solutions in both utility maximization and audience/seed diversification. In the future, it would be interesting to devise more efficient versions of the proposed algorithms and to explore a comprehensive metric jointly evaluating spread and diversity.

[2] www.kaggle.com/carolzhangdc/imdb-5000-movie-dataset

REFERENCES

[1] C. Borgs, M. Brautbar, J. Chayes, and B. Lucier. Maximizing social influence in nearly optimal time. In *SODA'14*, pages 946–957, 2014.

[2] A. Borodin, H. C. Lee, and Y. Ye. Max-sum diversification, monotone submodular functions and dynamic updates. In *PODS'12*, pages 155–166, 2012.

[3] N. Buchbinder, M. Feldman, J. S. Naor, and R. Schwartz. Submodular maximization with cardinality constraints. In *SODA'14*, pages 1433–1452, 2014.

[4] J. Carbonell and J. Goldstein. The use of mmr, diversity-based reranking for reordering documents and producing summaries. In *SIGIR'98*, pages 335–336, 1998.

[5] W. Chen, T. Lin, Z. Tan, M. Zhao, and X. Zhou. Robust influence maximization. In *KDD'16*, pages 795–804, 2016.

[6] W. Chen, C. Wang, and Y. Wang. Scalable influence maximization for prevalent viral marketing in large-scale social networks. In *KDD'10*, pages 1029–1038, 2010.

[7] W. Chen, Y. Yuan, and L. Zhang. Scalable influence maximization in social networks under the linear threshold model. In *ICDM'10*, pages 88–97, 2010.

[8] S. Gollapudi and A. Sharma. An axiomatic approach for result diversification. In *WWW'09*, pages 381–390, 2009.

[9] J. He, H. Tong, Q. Mei, and B. Szymanski. Gender: A generic diversified ranking algorithm. In *NIPS'12*, pages 1142–1150, 2012.

[10] D. Kempe, J. Kleinberg, and É. Tardos. Maximizing the spread of influence through a social network. In *KDD'03*, pages 137–146, 2003.

[11] Y. Lin, W. Chen, and J. C. Lui. Boosting information spread: An algorithmic approach. In *ICDE'17*, pages 883–894. IEEE, 2017.

[12] W. Lu, W. Chen, and L. V. Lakshmanan. From competition to complementarity: comparative influence diffusion and maximization. *PVLDB*, 9(2):60–71, 2015.

[13] M. McPherson, L. Smith-Lovin, and J. M. Cook. Birds of a feather: Homophily in social networks. *Annual review of sociology*, 27(1):415–444, 2001.

[14] Q. Mei, J. Guo, and D. Radev. Divrank: the interplay of prestige and diversity in information networks. In *KDD'10*, pages 1009–1018, 2010.

[15] W. W. Powell, D. R. White, K. W. Koput, and J. Owen-Smith. Network dynamics and field evolution: The growth of interorganizational collaboration in the life sciences. *American journal of sociology*, 110(4):1132–1205, 2005.

[16] M. Richardson, R. Agrawal, and P. Domingos. Trust management for the semantic web. In *ISWC'03*, pages 351–368, 2003.

[17] Y. Sun, J. Han, X. Yan, P. S. Yu, and T. Wu. Pathsim: Meta path-based top-k similarity search in heterogeneous information networks. *PVLDB*, 4(11):992–1003, 2011.

[18] F. Tang, Q. Liu, H. Zhu, E. Chen, and F. Zhu. Diversified social influence maximization. In *ASONAM'14*, pages 455–459, 2014.

[19] J. Tang, M. Qu, M. Wang, M. Zhang, J. Yan, and Q. Mei. Line: Large-scale information network embedding. In *WWW'15*, pages 1067–1077, 2015.

[20] Y. Tang, Y. Shi, and X. Xiao. Influence maximization in near-linear time: A martingale approach. In *SIGMOD'15*, pages 1539–1554, 2015.

[21] Y. Tang, X. Xiao, and Y. Shi. Influence maximization: Near-optimal time complexity meets practical efficiency. In *SIGMOD'14*, pages 75–86, 2014.

[22] H. R. Varian. *Intermediate Microeconomics: A Modern Approach*. WW Norton & Company, 2014.

[23] J. Yang and J. Leskovec. Overlapping community detection at scale: a nonnegative matrix factorization approach. In *WSDM'13*, pages 587–596, 2013.

[24] Y. Zhang. Diversifying seeds and audience in social influence maximization. *arXiv preprint arXiv:1906.09357*, 2019.

[25] X. Zhu, A. Goldberg, J. Van Gael, and D. Andrzejewski. Improving diversity in ranking using absorbing random walks. In *HLT-NAACL'07*, pages 97–104, 2007.